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ORIGINAL ARTICLE

Isolation and characterization of a subtype G2c variant of porcine epidemic diarrhea virus that adapts well to cell culture

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Abstract

Porcine epidemic diarrhea virus (PEDV) causes the third most important disease in the pig industry, after African swine fever and porcine reproductive and respiratory syndrome, and leads to illness or death of the entire litter, causing significant economic losses. In this study, three PEDV strains (HN-1, HN-2, and SC2023) were isolated from swine farms with suspected PEDV infections in Sichuan and Henan provinces. Phylogenetic analysis based on complete S gene sequences showed that all three strains belonged to the G2c subgroup. HN-1 adapted readily to cell culture, grew to a viral titer as high as 2×10^8 $TCID_{50}$/mL in Vero cells, and caused the formation of large syncytia. We analyzed the amino acid sequence of the HN-1 isolate and found that its S1 subunit contained a three-amino-acid insertion (^{355}KRL^{358}). A seven-amino-acid-deletion (^{1377}FEKVHVQ^{1383}) in the S2 subunit resulted in the partial deletion of the endocytosis signal YxxΦ and the complete deletion of the endoplasmic reticulum retrieval signal (ERRS) KVHVQ in the cytoplasmic tail of the S protein. Consequently, HN-1 is predicted to be less pathogenic than its parent strain, an attribute that facilitates rapid cell-to-cell spread by enhancing syncytium formation. In addition, strain HN-1 was found to have the mutation $^{884–885}$SG→RR, which may favor adaptation to cell culture by providing new trypsin cleavage sites. These results suggest that HN-1 is a G2c subtype variant that adapts well to cell culture and can be used to study the adaptive mechanisms of PEDV and develop attenuated vaccines.

Introduction

Porcine epidemic diarrhea (PED) is a highly contagious infectious disease characterized by acute inflammation, diarrhea, vomiting, and dehydration that is caused by porcine epidemic diarrhea virus (PEDV) [1]. PEDV is a single-stranded positive-stranded RNA virus belonging to the family *Coronaviridae* and the genus *Alphacoronavirus*. PEDV particles are globular in shape, with a distinct external envelope and peg- and petal-like spikes approximately 20 nm in length, similar to those of other members of the family *Coronaviridae* [2]. The PEDV genome is about 28 kb in length, containing 5' and 3' untranslated regions (UTRs) and seven open reading frames (ORFs), encoding four structural proteins – spike (S), envelope (E), membrane (M), and nucleocapsid (N) – as well as the pp1a, pp1ab, and ORF3 proteins, in the order 5'UTR-ORF1a/1b-S-ORF3-E-M-N-3'UTR [3].

The viral spike proteins, located on the surface of the virion, are type I transmembrane glycoproteins consisting of 1383 amino acids (aa) with a molecular weight of approximately 150 kDa [4]. The S proteins are responsible for membrane fusion activity during virus entry [5]. Fusion of the viral membrane with the host membrane is driven by conformational changes in the spike proteins, induced by a combination of receptor binding, protease cleavage, and a low-pH environment [6]. To establish efficient infection in host cells, coronaviruses have evolved the ability to utilize multiple proteases with different binding motifs

Min Su and Yi Yan contributed equally to this work.

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Nucleotide sequence data reported are available in the DDBJ/EMBL/GenBank databases under the accession number OR684450-OR684452.

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and cleavage efficiencies, including trypsin, chymotrypsin, transmembrane serine proteases, and cathepsin [2]. The S protein is cleaved by trypsin into two subunits, S1 (aa 1–789) and S2 (aa 790–1383), which induces changes in protein conformation. The S1 subunit specifically recognizes and binds to target cells and is the primary target of neutralizing antibodies [7]. The S2 subunit contains the fusion peptide, which interacts with the target membrane during virus entry and is responsible for trypsin-enhanced viral infection in cell culture [13]. The amino acid sequence of the S protein can be used to analyze the genetic relationship between different strains of PEDV and distinguish highly virulent from less-virulent strains, which is important for the development of PEDV vaccines.

The first PED epidemic was first reported by Belgian scientists and was caused by the prototype strain CV777 [8, 10]. Pigs of all ages were found to be susceptible to infection, but the severity of their symptoms varied. Piglets were the most susceptible at seven days of age, and mortality was as high as 100% [9]. Scientists in China reported the first PED outbreak in the country in 1973, but the virus was not isolated and characterized until 1984 [11]. In December 2010, a PEDV variant with enhanced pathogenicity appeared in China and continued to appear in North America and Asia, causing huge economic losses to the global pig industry [12]. These outbreaks were found to be caused by a PEDV variant with a low degree of similarity to the classical strain CV777. Phylogenetic analysis showed that the variant was a new subtype (PEDV GII) [13]. Li et al. [14] collected 882 diarrheal samples from 303 farms in China between 2017 and 2021 to investigate the epidemiology of PEDV. Sequencing and phylogenetic analysis of the S1 coding region of 104 strains revealed that 71 of the isolates (68.27%) belonged to the major G2c subgroup, whereas strains belonging to a new subgroup, G2d (9.62%), were identified in three provinces of China. Thus, G2c strains were primarily responsible for the epidemic in China from 2017 to 2021. These studies suggested that, during that time, genetic variation in endemic PEDV strains was accompanied by an increase in their severity and prevalence. Studies on the genetic diversity and molecular characteristics of PEDV are therefore urgently needed.

In this study, three PEDV G2c strains were isolated from samples from diseased pigs collected in Henan and Sichuan provinces in China. The complete genome and S gene sequences of these isolates were determined and analyzed. We found that one of them, isolate HN-1, adapted well to growth in cell culture, with a viral titer as high as 2×10^8 50% tissue culture infectious dose ($TCID_{50}$)/mL in Vero cells, resulting in the formation of large syncytia.

Materials and methods

Sample collection

Intestinal tissue and fecal samples of infected piglets were collected during diarrhea outbreaks on pig farms in Henan and Sichuan provinces of China, and reverse transcription polymerase chain reaction (RT-PCR) based on the PEDV M gene was used for detection of PEDV. PEDV-positive samples were weighed, added to Dulbecco's modified Eagle medium (DMEM) at a ratio of 1:5, and ground using a tissue grinder. The suspension was frozen and thawed three times, centrifuged at 12,000 × *g* for 5 min at 4°C, passed through a 0.22-µm filter, and stored at -80°C for subsequent virus isolation.

PEDV N-gene-based RT-PCR

Viral RNA was extracted from 200 µL of small intestine homogenate using an RNA extraction kit (SevenFast, China) according to the manufacturer's instructions. Based on genome sequences of classical and mutant PEDV strains in the NCBI GenBank database, a pair of PCR primers was designed for detection of a conserved region of the PEDV N gene. The sequence of the forward primer was 5'-GTCT TACATGCGAATTGACC-3', and that of the reverse primer was 5'-GGCATAGAGAGATAATGGCA-3'. cDNA was synthesized using a one-step RT-PCR kit (TaKaRa), and this cDNA was used as a template for PCR using the above N-gene-specific primers.

Virus isolation and propagation

Vero cells (ATCC CCL-81) were cultured in T25 culture flasks in DMEM containing 10% fetal bovine serum (Excell Bio, Australia) and 100 U of penicillin–streptomycin (Solarbio, China) per mL. When the cells reached approximately 80% confluence, they were washed twice with sterile phosphate-buffered saline (PBS, pH 7.2), and 1 mL of the filtered sample and 10 µg of trypsin (Gibco, USA) per mL were added to the Vero cells. After incubation at 37°C for 1 h, the cells were washed twice with PBS, 5 mL of maintenance medium (DMEM containing 10 µg of trypsin per mL and 1% penicillin–streptomycin) was added, and the inoculated cells were cultured in an incubator at 37°C with 5% CO_2. When 80–90% of the cells exhibited a cytopathic effect (CPE), the supernatant and cells were collected by freezing and thawing three times and then centrifuged at 1500 × *g* at 4°C for 15 min. The supernatant was collected and stored at -80°C.

Indirect immunofluorescence assay

Vero cells were digested, inoculated into 24-well cell culture plates, and infected when they formed a monolayer. After 24 h, the supernatant was discarded, and the cells were fixed with cold 4% paraformaldehyde for 30 min at room temperature and washed three times with PBS. The cells were permeabilized with 0.05% Triton X-100 for 10 min at room temperature and washed three times with PBS. Subsequently, 1% bovine serum albumin (BSA) and a murine anti-PEDV-N monoclonal antibody (mAb) were added. After incubation overnight at 4°C, the cells were washed three times with PBS. FITC-labeled sheep anti-mouse IgG (Boster Bio, Wuhan, China) was added, and after incubation at 37°C for 1 h, the cells were washed again with PBS, stained with 4′,6-diamidino-2-phenylindole (DAPI), and viewed using a fluorescence microscope (Leica, Germany).

$TCID_{50}$ assay

Vero cells were treated with trypsin, seeded in 96-well plates at a density of 5×10^4 cells per well, and incubated at 37°C for 24 h. The medium was removed, and 100 µL of tenfold serial dilutions of the virus was added to each well. The cells were observed under a microscope every 12 h for 5–7 days after infection to monitor the development of CPE, and the Reed–Muench method was used to calculate the virus titer as the $TCID_{50}$.

Plaque assay and growth curve

Vero cells were seeded into 12-well plates at a density of 5×10^5 cells/well. After 24 h of incubation at 37°C, the medium was removed by aspiration, and the cells were infected with 200 µL of tenfold serial dilutions of the viral suspension. After 1 h of incubation at 37°C, the medium was replaced with 1 mL of Eagle's minimum essential medium containing 1% agarose, 0.3% tryptone phosphate broth, and 100 U of penicillin-streptomycin and 10 µg of trypsin per mL. The plates were incubated at 4°C for 10 min to allow the overlying medium to solidify. The cells were then incubated at 37°C under 5% CO_2 to allow the formation of viral plaques. After 1 day of incubation, the cells were fixed in 10% (v/v) formaldehyde for 30 min, and viral plaques were observed by staining with 0.05% (w/v) crystal violet.

Viral growth curves of the PEDV isolates HN-1/2023, HN-2/2023, and SC2023 on Vero cells were constructed based on $TCID_{50}$ measurements. The cells were seeded in 6-well plates at a density of 10^6 cells per well, 2 mL of medium was added, and the cells were incubated at 37°C with 5% CO_2 for 24 h. The medium was then removed, and the viruses were inoculated with the virus suspension at an MOI of 0.01. At 12, 18, 24, 30, and 36 post-inoculation, the viral titer was determined using the $TCID_{50}$ method [15]. After washing twice with PBS, tenfold serial dilutions of the virus suspensions (10^{-1} to 10^{-8}) were added to eight wells containing monolayers of Vero cells in 96-well plates. The plates were incubated at 37°C for 5 days, and the viral titer at each time point was determined by microscopic observation of CPE.

Genome sequencing and phylogenetic analysis

Twenty-three primer pairs (Supplementary Table S1) were designed to amplify the entire genome of PEDV HN-1, HN-2, and SC2023 by conventional RT-PCR. The RT-PCR amplification procedure was as follows: pre-denaturation at 98°C for 30 s, followed by 30 cycles of denaturation at 98°C for 10 s, annealing at 55°C for 15 s, and extension at 72°C for 60 s. The RT-PCR product of each gene fragment was sequenced by Sangon Biotech (Shanghai, China). Complete genome sequences were assembled from the partial sequences and submitted to the GenBank database (https://www.ncbi.nlm.nih.gov/genbank/). The complete genome sequences and S gene sequences of HN-1, HN-2, and SC2023 were compared to those of 33 PEDV reference strains (Supplementary Table S2), using Clustal W in MEGA 11 software. Phylogenetic trees based on the whole-genome and S gene sequences were constructed by the neighbor-joining method. The final phylogenetic tree was drawn using TVBOT (Tree Visualization By One Table; https://www.chiplot.online/tvbot.html).

S protein homology modeling

The S proteins of PEDV strains HNAY and HN-1 were modeled using the SWISS-MODEL server with the CV777 S protein (PDB accession code 6u7k) as a template. The figures were created using PymoL.

Recombination analysis

Recombination events were detected using two methods. First, whole-genome sequences were compared and analyzed using the Recombination Detection Program (RDP v.4.1), which detects potential recombination events using eight algorithms (RDP, GENECONV, BootScan, MaxChi, Chimaera, SiScan, 3Seq, and LARD) [24]. Potential recombination events and breakpoints were further validated by similarity plot analysis using Simplot version 3.5.1.

Results

Virus isolation and identification

Total RNA was extracted from samples collected from diseased pigs in Sichuan and Henan provinces, and PEDV was identified by RT-PCR (Fig. 1A). The PEDV-positive samples were used for viral isolation. After inoculation with these samples, Vero (ATCC-CCL18) cells showed obvious CPE at 24 h, including rounding of cells, aggregation of nuclei into clusters, and appearance of syncytia (Fig. 1B). An immunofluorescence assay using an mAb against the PEDV N protein indicated that 50–60% of the cells were infected (Fig. 1C). Three PEDV strains were isolated and named "HN-1", "HN-2", and "SC2023".

In vitro characterization of PEDV

In a direct agarose overlay plaque assay, the HN-1, HN-2, and SC2023 isolates formed plaques, with HN-1 forming significantly larger plaques than the other two isolates, consistent with the observation that the HN-1 strain caused more-pronounced CPE (Fig. 2B). The viral titers of HN-1, HN-2, and SC2023 in Vero cells were 2×10^8 $TCID_{50}$/mL, 3.2×10^6 $TCID_{50}$/mL, and 7.67×10^4 $TCID_{50}$/mL, respectively. After inoculation of Vero cells at an MOI of 0.01, the titers of HN-1and HN-2 reached a maximum (8.18×10^7 $TCID_{50}$/mL and 1.22×10^6 $TCID_{50}$/mL) at 24 hpi, whereas the peak virus titer of SC2023 was 6.92×10^4 $TCID_{50}$/mL at 30 hpi (Fig. 2A).

Genomic and amino acid sequence analysis

The genome sequences of the three isolates (GenBank accession no. OR684450-OR684452) were obtained by combining the sequences of multiple PCR fragments (Fig. 3A-C) and found to be 28,008–28,018 nt in length. Comparison with the genome sequences of 33 reference strains obtained from the GenBank database (Supplementary Table S1) showed that the HN-1 isolate was 96-99.5% identical to the PEDV reference strains, 96-96.7% identical to the G1 isolates, 98.2–98.5% identical to the S-INDEL isolates, 98.5–98.9% identical to the G2a isolates, which are most common in China, and 98.4–99.8% identical to isolates of the new subtype G2c. HN-1 had the highest sequence identity (99.8%) to isolate CHN-SC2021 of subtype G2c (Supplementary Table S3).

The S gene of HN-1 is 4149 nt in length and encodes a 1383-aa protein. A multiple sequence alignment showed this protein to be 97.7–99.5% identical to those of the G2c isolates, 97.7–98.3% identical to those of the G2a isolates, 96.8–97% identical to those of the G2b isolates, 92.5–97% identical to those of the G1 isolates, and 95.6–95.7% identical to those of the S-INDEL isolates. HN-1 exhibited 98.3%

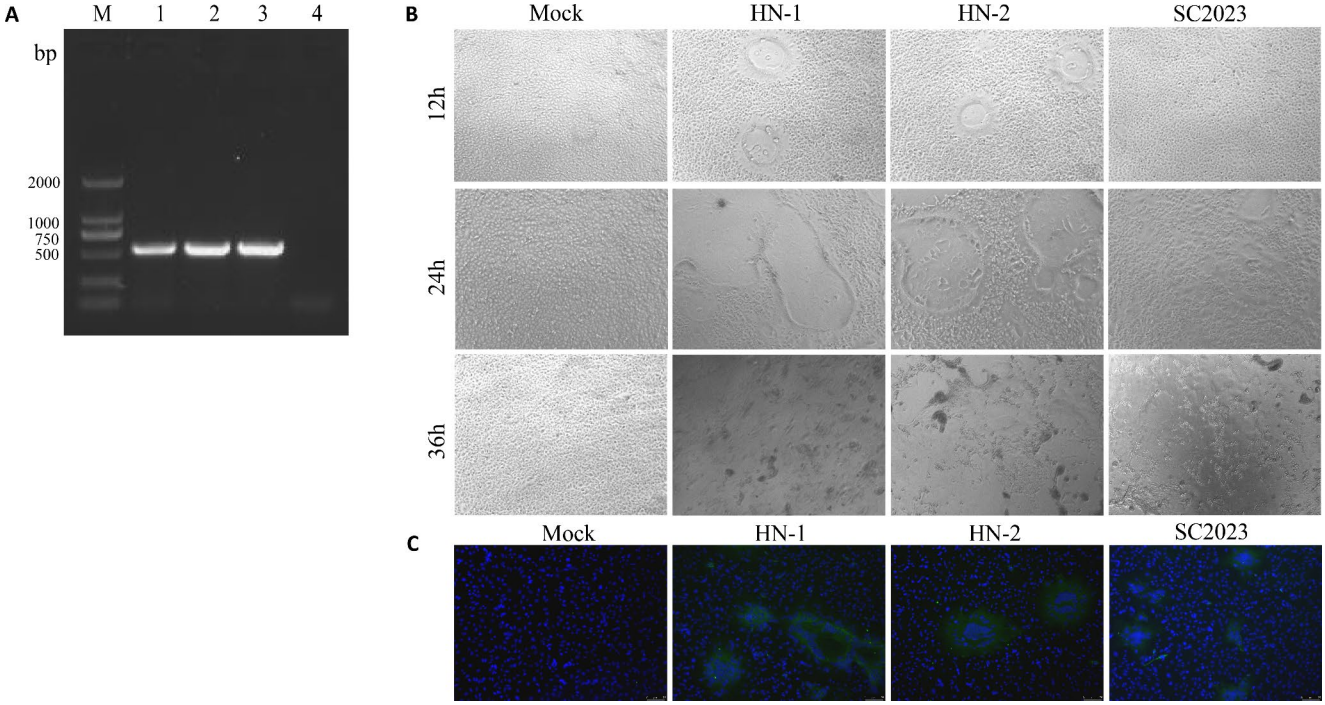

Fig. 1 Isolation and identification of PEDV strains. (**A**) Identification of PEDV in clinical samples by RT-PCR, using PEDV N-gene-specific primers. M, DL2000; 1–3, positive samples; 4, negative control. (**B**) Infection of Vero cells by PEDV strain s. HN-1 and HN-2 produced significant CPE at 24 h postinfection, and SC2023 produced significant CPE at 36 h postinfection. (**C**) IFA detection of PEDV antigens in infected Vero cells at 24 h postinfection

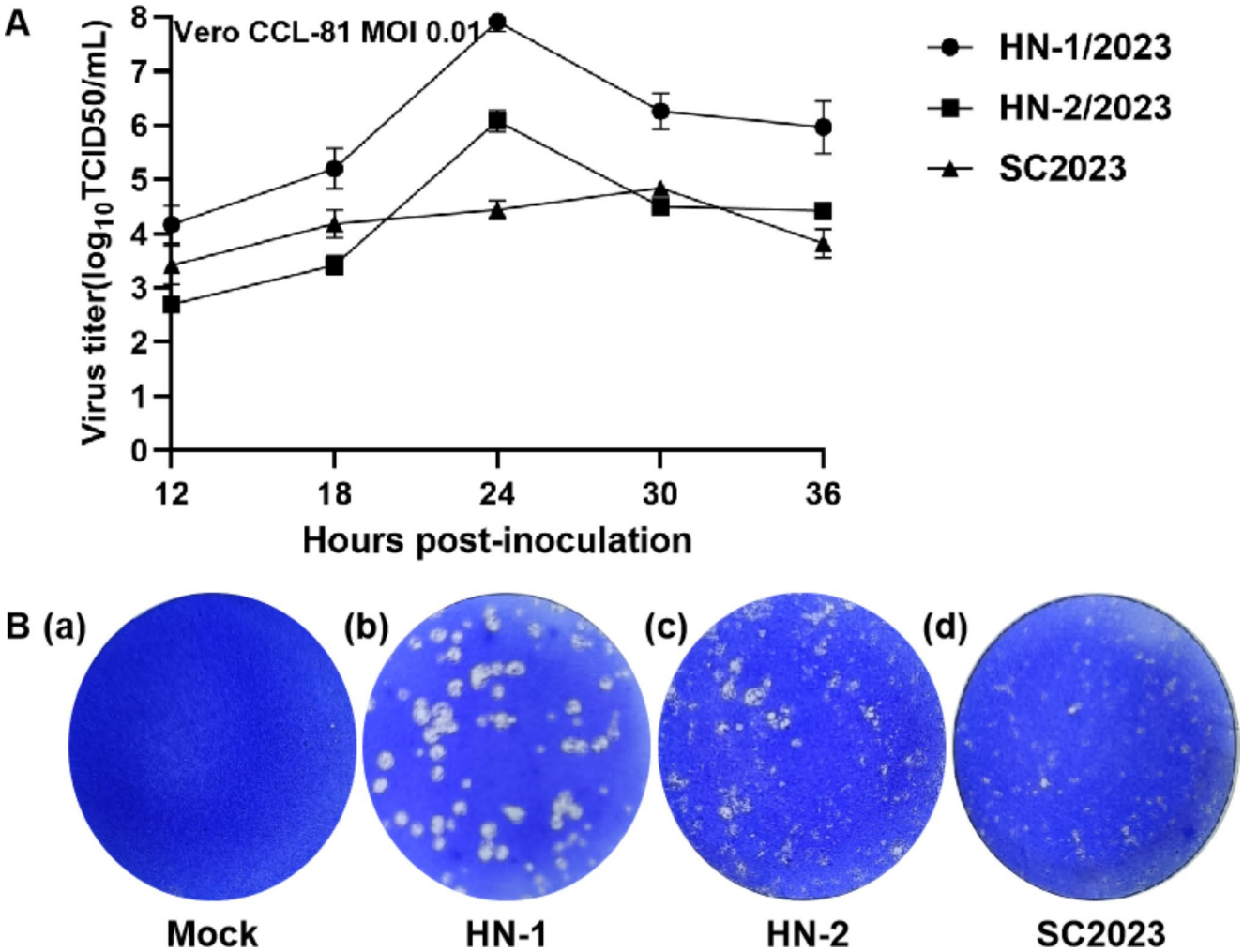

Fig. 2 *In vitro* characterization of PEDV isolates. (**A**) Multi-step growth curve of PEDV isolates on Vero cells inoculated at an MOI of 0.01. (**B**) Plaque morphology of PEDV isolates HN-1 (**b**), HN-2(**c**), and SC2023 (**d**) grown in Vero cells. Uninfected cells were used as a negative control (**a**)

and 98.4% sequence identity to HN-2 and SC2023, respectively (Supplementary Table S4).

Currently, six epitopes have been identified in S proteins: S10 (aa 19–220), S1A (aa 435–485), and the core neutralizing epitope (COE) on the S1 subunit (aa 499–638) and SS2 (aa 748–755), SS6 (aa 764–771), and 2C10 (aa 1368–1374) on the S2 subunit [16]. Compared with strain CV777, the three PEDV strains had nine mutations in common in the COE region (S516A, H521L, G523S, I527V, S549T, S594G, E609A, F612L, and V635I). In addition, the HN-1 strain contained two unique mutations (P60T and V609G) in the neutralizing epitopes S10 and COE. The three PEDV strains contained one mutation in the SS6 region (S766Y) and no mutations in the SS2 or 2C10 region (Fig. 4C).

The S1 subunit can be further subdivided into six structural domains: the signal peptide (SP, aa 1–18), structural domain 0 (aa 19–219), structural domain A (aa 220–509), structural regions B (aa 510–639), structural domain C, and structural domain D (aa 640–728) [17]. Sequence analysis demonstrated that, compared with the classical strain CV777, the HN-1 strain has a novel 9-nt insertion at nt 1,063 – 1,072 in the S1A region (Fig. 4A), resulting in a 3-aa insertion at aa 355–358 in structural domain A (Fig. 4B). The HN-1 and HNAY2016 S protein structures were predicted using SWISS-MODEL based on the structure of the S protein of strain CV777 (PDB accession number 6u7k) (Fig. 5B), and this showed the amino acid insertion to be located in a flexible loop (Leu354–Ala363) at the top of the trimer [18]. A specific double mutation was found in the S2 subunit ($^{884–885}$SG→RR), and a deletion of nt 4130–4131 of the S gene resulted in premature termination of S protein translation, resulting in the absence of the last seven amino acids of the S2 subunit at positions 1377–1383 (Fig. 4D).

Phylogenetic and recombination analysis

Phylogenetic trees based on complete genome sequences showed that the 36 PEDV strains analyzed were divided into

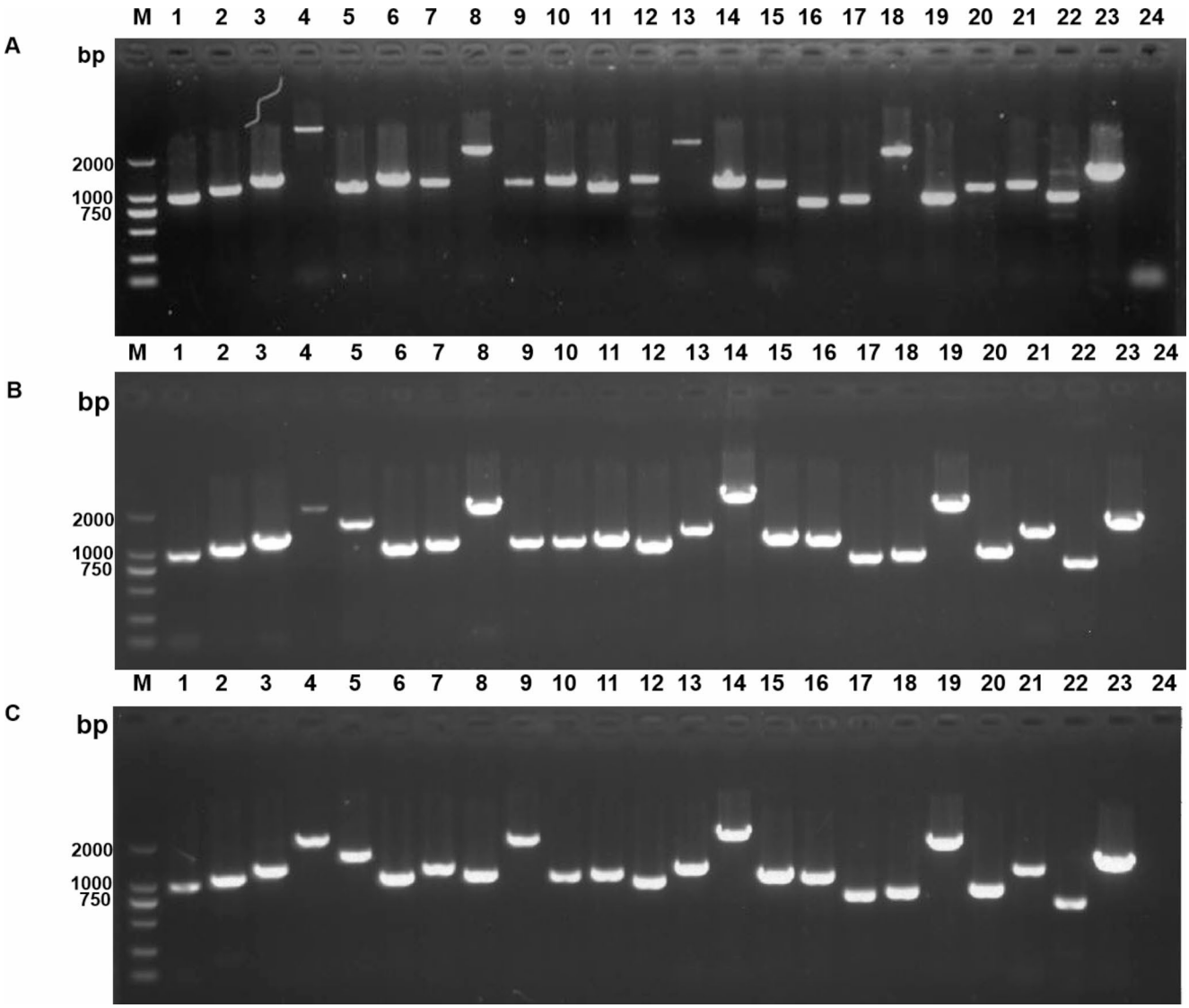

Fig. 3 Gel electrophoresis results showing amplification of the entire PEDV genome by PCR. The viral genome was amplified in 23 fragments, the PCR products were sequenced, and the sequences were combined to assemble the complete genome sequence (**A**) Isolate HN-1. (**B**) Isolate HN-2. (**C**) Isolate SC2023. M, DL2000; 1–23, 23 fragments of the viral genome; 24, negative control

two branches: G1 (classical strains) and G2 (variant strains). The G2 branch was further divided into three subgroups – G2a, G2b, and G2c – and the three PEDV isolates from this study belonged to the G2c subgroup (Fig. 6A). Another tree based on S gene sequences gave similar results (Fig. 6B).

To look for evidence of recombination in the three new PEDV isolates, their sequences were compared to those of the reference strains of the subtypes Gla, G1b, S-INDEL, G2a, G2b, and G2c using RDP4. A possible recombination event was identified in the HN-2 isolate with putative breakpoints located at nt 182 and 980 in ORF1a. The major parental strain was identified as HN-1 of the G2c subgroup, and the minor parental strain was identified as SXSL of the G2c subgroup (Fig. 7A).

SimPlot analysis demonstrated that the entire genome of HN-2 was very similar to those of HN-1 and SXSL. Consistent with the results of RDP4 analysis, a potential recombination breakpoint was detected in ORF1a of HN-2 (Fig. 7B). In summary, no recombination events were identified in HN-1 or SC2023, whereas the HN-2 genome shows evidence of recombination within the G2c subgroup.

Discussion

PEDV is a major threat to the global pig industry. Its genome is a single positive-strand RNA that is highly susceptible to mutations, with large differences between strains of different genotypes, resulting in a complex epidemiological

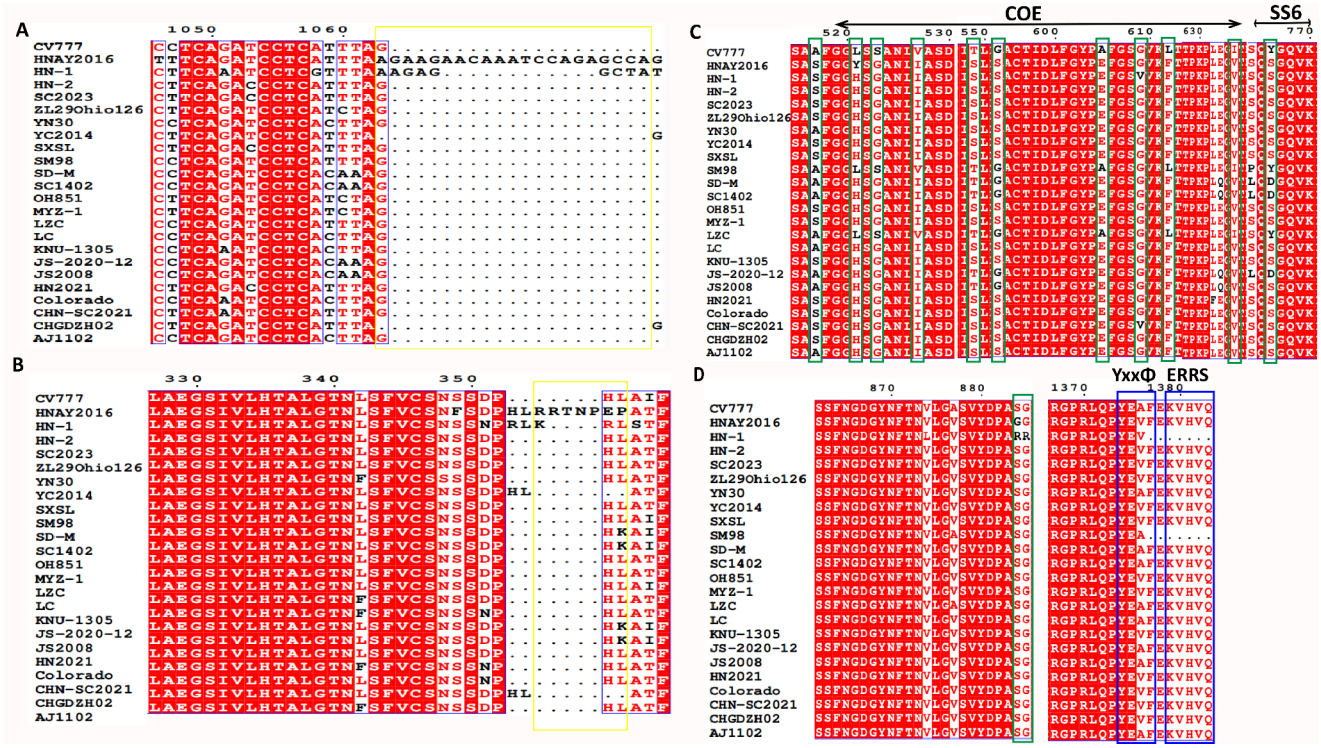

Fig. 4 Analysis of amino acid mutations in the S protein of virulent PEDV strains. The amino acid sequences of the S proteins of 24 PEDV strains were compared using the CLUSTAL W method. (**A**) Insertion of nine nucleotides in the region encoding the S1A domain (AAGAG GCTA). (**B**) Insertion of three amino acids (RLS) in the S1A domain (residues 220 to 509). (**C**) Amino acid mutations in the neutralization epitopes of the S protein (S10, residues 19–220; COE, residues 499–638; SS6, residues 764–771; 2C10, residues 1368–1374). (**D**) HN-1 strain $^{884–885}$SG→ RR mutation, partial deletion of YxxΦ, and complete deletion of the endoplasmic reticulum retrieval signal (KVHVQ). Mutations, insertions, and deletions are highlighted in green, yellow, and blue, respectively

situation [19]. Early outbreaks of disease caused by PEDV led to a low rate of piglet mortality, but in 2010, an outbreak in China caused by G2-type PEDV resulted in piglet mortality rates up to 100% [9] due to a change in the pathogenicity of the virus. Recently, a PEDV variant with a truncated S protein appeared in Japan, and this virus showed tropism for airway epithelial cells, submucosal glands, and biliary epithelial cells [20, 21]. In addition to changes in biological properties, genetic variations in the virus remain the main cause of immunization failure. With the continuous development of detection technologies and bioinformatics, an increasing number of PEDV strains have been identified. Monitoring of circulating PEDV strains and the development of updated vaccines are key to the prevention and control of PEDV.

In this study, three PEDV strains – HN-1, HN-2, and SC2023 – were isolated from samples collected from diseased pigs in Sichuan and Henan provinces of China, and their genome sequences were determined and analyzed. The HN-1 isolate had the highest sequence similarity (99.8% identity) to strain CHN-SC2021 of subgroup G2c. The HN-1 isolate was found to have 96.3% and 98.3% genome-wide sequence identity to the vaccine strains CV777 and AJ1102, respectively, which occupy different branches of the phylogenetic tree. The S gene of HN-1 was found to be 93.3% and 97.0% identical, respectively, to those of the vaccine strains. Currently, PEDV G2 strains are divided into three subgroups: G2a, G2b, and G2c. The G2a type mainly contains the epidemic mutant strains from 2010, the G2b subgroup includes epidemic mutant strains from the United States from 2013, and the G2c subgroup includes epidemic mutant strains from China from 2015 [22, 23]. Phylogenetic analysis based on complete genome sequences and S gene sequences showed that all three PEDV isolates from this study belong to the G2c subgroup.

The S gene of the HN-1 isolate was found to contain multiple mutations in the S10 structural domain, including one unique mutation (P60T), compared to that of the vaccine strain CV777. The strains PC220A-P120, PT-P96, YN144, FJzz1-F200, and OH851 contain mutations in the S10 structural domain, which binds to host cell ?? sialic ?? acid receptors attached to virus particles [24]. Mutations in this structural domain may disrupt binding to ?? sialic ??-acid-containing glycans, thereby reducing the efficiency of entry of the virus into cells. The HN-1 strain showed a 3-aa (aa 355–358) insertion in the PEDV S1A structural domain.

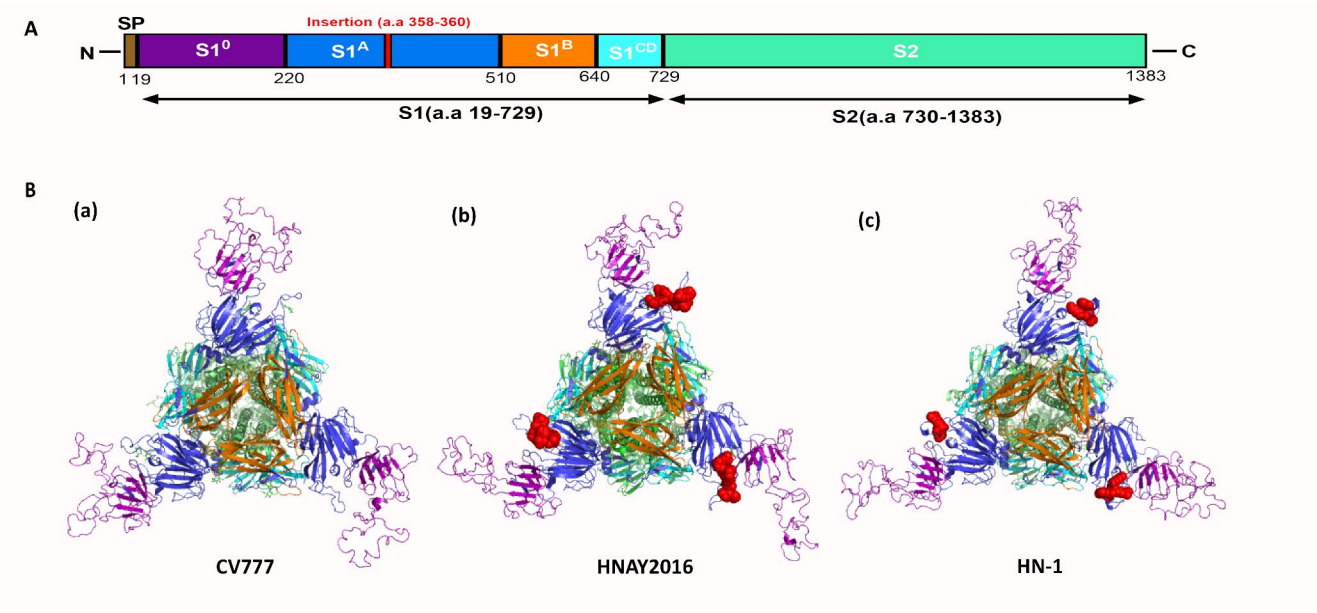

Fig. 5 Homology modeling of the S protein. (**A**) Schematic representation of the PEDV S protein. The S protein is divided into an S1 subunit and an S2 subunit. The S1 subunit is further divided into four structural domains: structural domain 0 (aa 19–219, purple), structural domain A (aa 220–509, blue), structural domain B (aa 510–639, orange), and structural domain CD (aa 640–728, arctic blue). The S protein of isolate HN-1 has an insertion of three amino acids in structural domain A (aa 358–360, red). The S2 subunit (aa 730–1383) is colored lime green. (**B**) The CV777 S protein crystal structure was used as a template for modeling the S proteins of PEDV strains HNAY and HN-1 using SWISS-MODEL. In the models, S10, S1A, S1B, S1CD, and S2 are shown in purple, blue, orange, cyan, and lime green, respectively. The 7-aa insertion is shown as a red dot in the HNAY structural model, and the 3-aa insertion is shown as a red dot in the HN-1 structural model

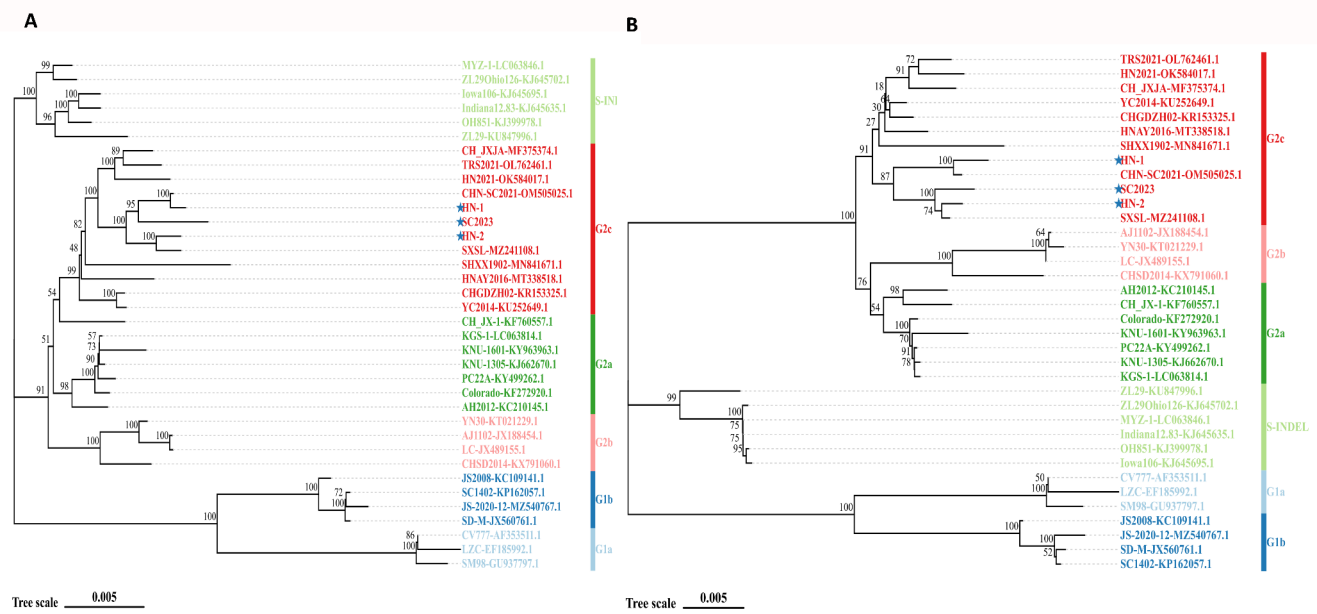

Fig. 6 Phylogenetic trees based on the entire genome of PEDV (**A**) and on the S gene (**B**), constructed by the neighbor-joining method using the MEGA 11 program. Scale bars indicate the number of substitutions per 20 amino acids. Bootstrap values for 1000 resamplings are shown at the nodes. The blue stars indicate the three PEDV strains isolated in this study. The analysis showed that the three PEDV strains were closely related to members of subgroup G2, and all three belonged to subgroup 2c

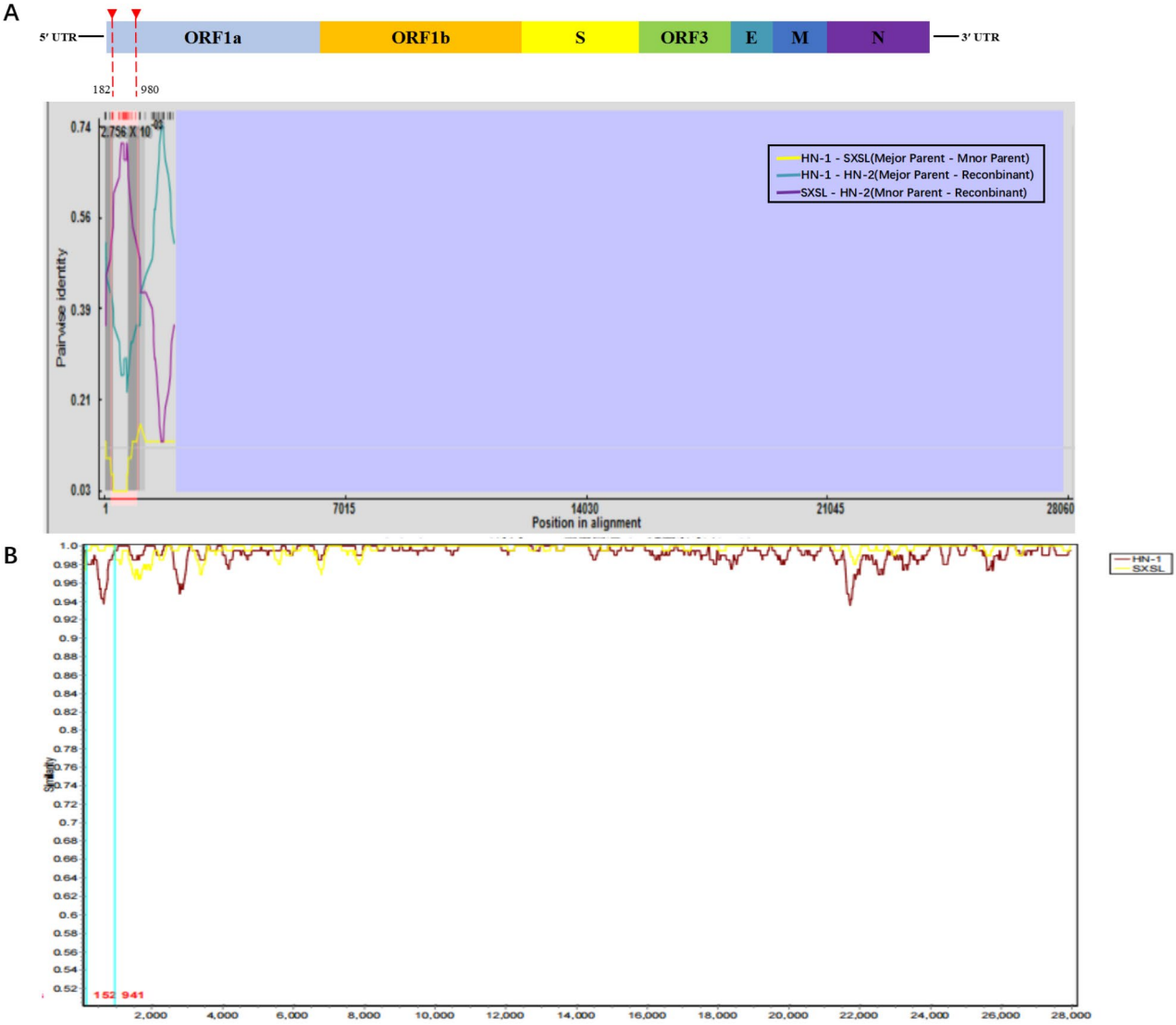

Fig. 7 Recombination breakpoints in the genome of the PEDV HN-2 isolate predicted by RDP4. The potential parental strains and the recombinant isolate are shown in teal (major), purple (minor) and yellow (recombinant), respectively. Red arrows indicate recombinant breakpoints. The y-axis shows the percentage of permutated trees employing a sliding window of 1000 bp and a step size of 20 bp. The other options including Kimura (2-parameter) distance model, 2.0 Ts/Tv ratio, neighbor-joining tree model and 1000 bootstrap replicates are used. The RDP4 results are further identified by using a Simplot analysis. The query sequence from HN-2 was used to compare with the parental sequences HN-1 and SXSL. The x-axis indicates entire genome sequences, and the y-axis represents the similarity value. The regions of recombinant breakpoint are shown within two red lines

This insertion is located in a flexible loop at the top of the trimer (aa 354–363). However, this region has not been resolved in a structural model of the PEDV S protein.

The COE (aa 499–638) and SS6 (^{764}LQDGGQVKI^{771}) epitopes are diverse in most field strains and are highly variable [25]. Variations in these epitopes may be important references for the development of weakly virulent vaccines. The HN-1 strain was found to have 10 mutations in the COE epitope, one of which was a unique mutation (V609G) that was not found in other strains. Mutations in the neutralization epitope may affect the efficacy of the vaccine, which may also explain why PEDV remains prevalent despite the availability of vaccines. No amino acid changes were identified in the SS2 epitope, whereas in the SS6 epitope, one specific amino acid mutation (Y766S) was identified. Deletions, insertions, and substitutions in the S gene may affect the structure of the protein, which can affect the virulence of the virus. However, whether the amino acid substitutions identified in this study affect the antigenicity and pathogenicity of PEDV has not been investigated.

The PEDV S protein contains two sorting motifs in the cytoplasmic tail: the endocytosis signal YxxΦ and the

endoplasmic reticulum retrieval signal KVHVQ. Deletion of these two motifs has been shown to reduce pathogenicity significantly in piglets [26]. Variants lacking these two motifs (FL2013, SM-98, CHM2013, PC22A-P120, and KNU-141112) exhibit reduced virulence in piglets [27–29]. The HN-1 isolate has a deletion of nt 4130–4131, resulting in a premature stop codon that causes the S protein to be truncated by seven amino acids (FEKVHVQ). This deletion results in the loss of the YxxΦ and KVHVQ motifs and is therefore likely to reduce the virulence of the virus in piglets. However, deletion of these two motifs is not the only determinant of reduced virulence, and therefore, this hypothesis needs to be verified by animal experiments.

The ability of PEDV to adapt to cell culture varies significantly among isolates [30]. The HN-1 isolate adapted readily to growth in Vero cells, as evidenced by infection titers up to 2×10^8 $TCID_{50}$/mL and the formation of large syncytia. Trypsin promotes cell fusion and viral infection by cleaving the S protein [31]. The S888R (R885) mutation might facilitate adaptation to cell culture by introducing new trypsin cleavage sites [32]. Since trypsin cleaves only at arginine and lysine residues, we hypothesize that the $^{884–885}$SG→RR mutation promotes adaptation to Vero cells by facilitating trypsin cleavage. The size of the syncytia induced by the virus is correlated with the number of S proteins on the cell surface [33]. Both the YxxΦ and KVHVQ motifs are involved in the regulation of the amount of PEDV S protein on the cell surface: the KVHVQ motif retains S proteins in the ERGIC and restricts the translocation of S proteins to the cell surface, whereas the YxxΦ motif reduces the number of surface S proteins through endocytosis [26]. The lack of both of these motifs in the HN-1 strain is likely to increase the number of S proteins on the cell surface, favoring the formation of large syncytia.

In conclusion, our findings indicate that PEDV subgroup G2c strains are endemic to Henan and Sichuan provinces and that there are multiple antigenic variants of the S protein with differences in their epitopes. The HN-1 strain is readily adaptable to growth in cultured cells and can be used to study the adaptive mechanisms of PEDV and develop attenuated vaccines.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Min Su and Yi Yan. The first draft of the manuscript was written by Min Su, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets generated during and/or analysed in the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors report no conflict of interest.

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